

CASE STUDY REPORT

Routing Protocol Failures

A Root-Cause Analysis of the March 2024 NimbusEdge Network Outage

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SUBJECT AREA

Computer Networks

Prepared using the Incident Report, Network Topology Diagram, RCA Report,
NOC Change Log and Client SLA Breach Notice

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1. Introduction

In March 2024, NimbusEdge Technologies experienced a major network outage that disrupted service for six enterprise clients for approximately nine hours. Investigation showed that the outage was not the result of a single fault, but of four separate routing-protocol issues occurring across different parts of the network: a hop-count limitation in RIP, a slow convergence response in IGRP, a configuration mismatch in OSPF, and a failed automatic failover in BGP.

This report analyses each failure individually — what happened, why it happened, why it was difficult to detect, and what corrective action is recommended — before drawing an overall conclusion about the network's design and operational practices.

Documents Analysed

- Incident Report
- Network Topology Diagram
- Root Cause Analysis (RCA) Report
- NOC Change Log
- Client SLA Breach Notice

2. Incident Overview

The table below summarises the four protocol failures, their location in the network, and the underlying cause of each.

Protocol	Location	Root Cause
RIP	Chennai HQ	Delivery-department route reached 16 hops and was marked unreachable
IGRP	Bengaluru – Pune – Hyderabad	Degraded link was not abandoned quickly enough (slow convergence)
OSPF	Core Backbone – Hyderabad	Incorrect Area ID prevented adjacency from forming
BGP	ISP-A / ISP-B Peering	Automatic failover to the backup ISP did not trigger as designed

Q1 RIP — Chennai HQ Campus

What happened?

Chennai HQ added a new Delivery-department subnet while the site was still running RIP v2. The route to this subnet reached a hop count of 16. Since RIP only treats routes of 1–15 hops as reachable, the Delivery subnet was silently marked unreachable. The fault surfaced only after client support tickets were raised, because standard monitoring did not flag the routing failure.

Why did 16 hops cause the failure?

RIP uses hop count — the number of routers a packet must cross — as its sole routing metric. Values of 1–15 are considered reachable; a metric of 16 represents infinity, i.e. unreachable. Once the Delivery route's hop count reached 16, RIP stopped advertising it as usable.

Why was it not automatically detected?

This was a logical routing failure rather than a physical link failure: the routers and interfaces involved continued to function normally, so link-level monitoring showed no problem. The RCA identified a genuine monitoring gap — route reachability was not tracked with the same rigour as physical link status — compounded by RIP's inherent 15-hop ceiling.

Recommended fix

Migrate Chennai HQ from RIP to OSPF. OSPF scales further and has no hop-count ceiling, making it a better fit as the site grows. RIP was a reasonable original choice for a small, stable campus because of its simplicity — the problem only emerged once the network expanded beyond that scope.

Rule of thumb: 1–15 hops = reachable | 16 hops = unreachable (RIP infinity metric).

Q2

IGRP — Bengaluru–Pune–Hyderabad Region

What happened?

An ISP-side fault degraded the Bengaluru–Pune fibre link: effective bandwidth dropped to roughly 2 Mbps and latency increased. A healthier alternate path existed via Pune–Hyderabad–Bengaluru, but IGRP took too long to shift traffic onto it, causing severe application latency for clients on the affected segment.

What does the IGRP metric show?

Classic IGRP computes a composite metric weighted mainly by bandwidth and delay. For a 2,000 kbps link, the bandwidth component works out to $10,000,000 \div 2,000 = 5,000$, before delay is added and final scaling applied. Because delay also rose, the composite metric for the degraded path should have worsened further, and the healthier alternate should have been preferred once its complete metric was lower.

Data limitation: the case materials do not provide complete delay values for both paths, so an exact side-by-side metric comparison would require assuming figures not present in the source documents.

Why was the problem mainly convergence?

IGRP's metric design was not itself at fault — using bandwidth and delay together is more informative than hop count alone. The real problem was speed of reaction: the network did not converge onto the healthier path quickly enough after conditions changed. In short, IGRP had the right information but acted on it too slowly.

Recommended fix

Migrate the regional network to OSPF. IGRP was originally an improvement over RIP because it factored in bandwidth and delay, but a growing regional network needs the faster convergence and stronger scalability that OSPF provides, alongside active monitoring of both link performance and routing convergence time.

Q3 OSPF — Core Backbone Migration

What happened?

NimbusEdge was migrating its core network to multi-area OSPF, with Hyderabad intended to sit in Area 3. The Hyderabad router was instead configured with the wrong Area ID. Because the ID did not match the design, the router could not form the expected OSPF adjacency — creating a routing black hole until a later change record corrected the Area ID.

Why does the Area ID matter?

OSPF neighbours must agree on several parameters before forming an adjacency: Area ID, Hello interval, Dead interval, authentication settings (where used), and compatible network types. In this incident the specific mismatch was the Area ID — because the two routers disagreed on which area they belonged to, the neighbour relationship could not form at all.

Why was it harder to detect?

A physical link failure is usually obvious, since the interface itself goes down. Here the physical connection stayed up while the routing layer silently failed, so basic link monitoring showed everything as healthy. The RCA also points to a change-management gap: the misconfiguration was not caught because post-change validation was insufficient.

How should it be fixed?

Correct the Hyderabad router's Area ID to the intended Area 3 assignment, then verify that the OSPF neighbour reaches FULL state, confirm route/LSA exchange, check the routing table, and run an end-to-end connectivity test. To prevent recurrence, introduce peer review of configuration changes, a formal change checklist, and mandatory post-change adjacency verification.

Why use multi-area OSPF?

OSPF offers faster convergence, better scalability, and detailed link-state visibility compared with the older RIP/IGRP design — making it the more appropriate protocol for a multi-site enterprise backbone.

Q4 BGP — ISP Peering and Failover

What happened?

NimbusEdge maintained two ISP connections: ISP-A as primary and ISP-B as backup. When ISP-A failed, BGP was expected to withdraw routes through ISP-A and automatically select ISP-B. This automatic failover did not occur, forcing engineers to intervene manually — a delay that contributed directly to the length of the outage and to the resulting SLA breach.

Key BGP attributes

Attribute	Meaning
LOCAL_PREF	Preference for an outbound path within the company's own AS — higher value is preferred.
AS-PATH	The sequence of autonomous systems a route has traversed; a shorter path is generally preferred when higher-priority attributes are equal.
MED	Influences the preferred entry point between two connected autonomous systems; a lower value is generally preferred.

The RCA points specifically to a LOCAL_PREF / AS-PATH policy problem as the reason automatic failover did not behave as expected.

Why is an untested BGP policy risky?

BGP governs communication with external autonomous systems and ISPs, so a policy mistake can affect a large volume of external traffic. The resulting chain of events was: ISP-A failed → automatic failover did not trigger → engineers intervened manually → recovery was delayed → clients were impacted → the SLA was breached.

What should the BGP policy look like?

During normal operation, ISP-A should carry a higher preference than ISP-B — for example, LOCAL_PREF 200 for ISP-A versus 100 for ISP-B (illustrative values; the key requirement is that ISP-A remains primary and ISP-B remains backup). When ISP-A's route is withdrawn, ISP-B should automatically become the best available path. NimbusEdge should also run regular, controlled failover tests to confirm this behaviour before it is needed in a real incident.

Before / After

Before	ISP-A failed, but the existing policy did not produce the expected automatic switch to ISP-B.
After	ISP-A is primary with higher LOCAL_PREF; ISP-B is backup with lower LOCAL_PREF. When ISP-A fails, its route is withdrawn and ISP-B becomes active automatically.

5. Overall Conclusion

The NimbusEdge outage was the product of four distinct routing failures rather than one isolated cause: RIP hit its inherent 15-hop limit, IGRP reacted too slowly to a degraded link, OSPF was undermined by an incorrect Area ID, and BGP failed to deliver the automatic ISP failover it was designed for.

The common thread across all four cases is that effective routing depends on more than protocol selection: it requires correct configuration, meaningful monitoring, disciplined change validation, and regular, realistic failure testing.

In simple terms: choose the right protocol for the network's scale, configure it correctly, monitor the actual routing state (not just link status), and test what happens when a link or path fails — before a real incident forces the test.

6. Documents Referenced

- Incident_Report_NimbusEdge_March2024.pdf
- Network_Topology_Diagram_NimbusEdge.pdf
- RCA_Report_Q1_2024.docx
- NOC_Change_Log_Feb-Mar_2024.xlsx
- Client_SLA_Breach_Notice.pdf